

Chapter 7: Eigenvalues and Eigenvectors

Chapter overview

Eigenvalues and eigenvectors identify directions that a matrix transforms only by scaling. This viewpoint leads to diagonalization, orthogonal diagonalization of symmetric matrices, stable population structures, decoupled differential equations, principal axes of quadratic forms, and constrained optimization.

Original contexts

All worked numerical examples are original. Themes selectively use Spurs-inspired values, hiking, badminton, travel, driving, Backstreet Boys, and R&B music. Fictional data are identified as teaching models.

1 Eigenvalues and Eigenvectors

Eigenvalue and eigenvector

For an $n \times n$ matrix A , a scalar λ is an eigenvalue if there is a nonzero vector $\mathbf{v}$ satisfying

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The vector $\mathbf{v}$ is an eigenvector corresponding to λ .

An eigenvector determines an invariant line. Positive eigenvalues preserve direction, negative eigenvalues reverse direction, and zero eigenvalues collapse the vector to zero.

Spurs-inspired verification

Let

$$A = \begin{bmatrix} 20 & 1 \\ 1 & 20 \end{bmatrix}.$$

Then

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 21 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 19 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Thus the eigendirections are $y = x$ and $y = -x$.

Eigenspace

$$E_\lambda = \ker(A - \lambda I).$$

It consists of all eigenvectors for λ together with the zero vector.

The eigenspace is a subspace

For each eigenvalue λ , the set E_λ is a subspace of $\mathbb{R}^n$.

Proof

First, $(A - \lambda I)\mathbf{0} = \mathbf{0}$, so $\mathbf{0} \in E_\lambda$. If $\mathbf{u}, \mathbf{v} \in E_\lambda$, then

$$(A - \lambda I)(\mathbf{u} + \mathbf{v}) = (A - \lambda I)\mathbf{u} + (A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Thus $\mathbf{u} + \mathbf{v} \in E_\lambda$. If c is a scalar, then

$$(A - \lambda I)(c\mathbf{u}) = c(A - \lambda I)\mathbf{u} = \mathbf{0},$$

so $c\mathbf{u} \in E_\lambda$. Therefore E_λ is a subspace. $\square$

Characteristic equation

A scalar λ is an eigenvalue of A if and only if

$$\det(A - \lambda I) = 0.$$

Proof

The equation $A\mathbf{v} = \lambda\mathbf{v}$ is equivalent to

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

A nonzero solution exists exactly when $A - \lambda I$ is singular, and a square matrix is singular exactly when its determinant is zero. $\square$

Detailed 2×2 example

For

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix},$$

$$\begin{aligned} \det(A - \lambda I) &= (4 - \lambda)(3 - \lambda) - 2 \\ &= \lambda^2 - 7\lambda + 10 \\ &= (\lambda - 5)(\lambda - 2). \end{aligned}$$

For $\lambda = 5$, solve $(A - 5I)\mathbf{v} = \mathbf{0}$:

$$A - 5I = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow -R_1} \begin{bmatrix} 1 & -2 \\ 1 & -2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

The remaining equation is $x - 2y = 0$, so $x = 2y$. Letting $y = t$ gives

$$\mathbf{v} = t \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad E_5 = \text{span}\{(2, 1)\}.$$

A direct check is

$$A \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

For $\lambda = 2$,

$$A - 2I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus $x + y = 0$, so $x = -y$. Letting $y = t$ gives

$$\mathbf{v} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad E_2 = \text{span}\{(-1, 1)\}.$$

Again,

$$A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Algebraic and geometric multiplicity

For

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix},$$

$\lambda = 2$ has algebraic multiplicity 3, but

$$B - 2I = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

implies $x_3 = 0$. Hence

$$E_2 = \text{span}\{(1, 0, 0), (0, 1, 0)\}$$

and the geometric multiplicity is 2.

Eigenvalues of triangular matrices

The eigenvalues of a triangular matrix are its diagonal entries.

Proof

For triangular A , the matrix $A - \lambda I$ is triangular, so

$$\det(A - \lambda I) = \prod_{j=1}^n (a_{jj} - \lambda).$$

The roots are the diagonal entries. $\square$

A polynomial transformation

For $T(p) = p + xp'$ on P_2 ,

$$T(a + bx + cx^2) = a + 2bx + 3cx^2.$$

Relative to $\{1, x, x^2\}$, the matrix is $\text{diag}(1, 2, 3)$, so the basis vectors are eigenvectors.

2 Diagonalization

Diagonalizable matrix

A is diagonalizable if there is an invertible matrix P such that

$$P^{-1}AP = D$$

with D diagonal.

Similar matrices have the same characteristic polynomial

If $B = P^{-1}AP$, then A and B have the same characteristic polynomial.

Proof

$$\lambda I - B = P^{-1}(\lambda I - A)P.$$

Therefore,

$$\det(\lambda I - B) = \det(P^{-1}) \det(\lambda I - A) \det(P) = \det(\lambda I - A).$$

□

Diagonalization criterion

An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.

Proof

If $\mathbf{p}_1, \dots, \mathbf{p}_n$ are independent eigenvectors, form $P = [\mathbf{p}_1 \cdots \mathbf{p}_n]$ and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then

$$AP = [A\mathbf{p}_1 \cdots A\mathbf{p}_n] = [\lambda_1\mathbf{p}_1 \cdots \lambda_n\mathbf{p}_n] = PD.$$

Since P is invertible, $P^{-1}AP = D$. Conversely, if $P^{-1}AP = D$, then $AP = PD$. Comparing columns shows that the columns of P are eigenvectors. Since P is invertible, they are independent. □

Distinct eigenvalues give independent eigenvectors

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Proof

Use induction. If $\sum_{j=1}^k c_j \mathbf{v}_j = 0$, apply A , then subtract λ_k times the original relation:

$$\sum_{j=1}^{k-1} c_j (\lambda_j - \lambda_k) \mathbf{v}_j = 0.$$

The induction hypothesis forces $c_1 = \cdots = c_{k-1} = 0$, and then $c_k = 0$. □

Complete 3×3 diagonalization

Let

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Its characteristic polynomial is

$$(\lambda - 1)(\lambda - 2)(\lambda - 3).$$

We now compute every eigenspace.

For $\lambda = 1$,

$$A - I = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Use

$$R_2 \leftarrow R_2 + R_1, \quad R_3 \leftarrow R_3 - R_1,$$

to obtain

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus $x_1 = -x_3$ and $x_2 = 0$, so

$$E_1 = \text{span}\{(1, 0, -1)\}.$$

For $\lambda = 2$,

$$A - 2I = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence $x_1 = -x_3$ and $x_2 = x_3$, so

$$E_2 = \text{span}\{(-1, 1, 1)\}.$$

For $\lambda = 3$,

$$A - 3I = \begin{bmatrix} -1 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & 1 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore $x_1 = -x_2$ and $x_3 = 0$, so

$$E_3 = \text{span}\{(-1, 1, 0)\}.$$

The three eigenvectors correspond to distinct eigenvalues, so they are linearly independent. Thus

$$P = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Since $\det(P) = -1$,

$$P^{-1} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{bmatrix}, \quad P^{-1}AP = D.$$

Therefore

$$A^k = PD^kP^{-1}.$$

For example,

$$A^5 = \begin{bmatrix} 212 & -31 & 211 \\ -211 & 32 & -211 \\ 31 & 31 & 32 \end{bmatrix}.$$

The Jordan-type matrix

$$J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

is not diagonalizable: the eigenvalue 2 has algebraic multiplicity 2 but eigenspace dimension 1.

3 Symmetric Matrices and Orthogonal Diagonalization

Symmetric and orthogonal matrices

A real matrix is symmetric when $A^T = A$. A real square matrix P is orthogonal when $P^T P = I$, equivalently $P^{-1} = P^T$.

Column criterion

P is orthogonal if and only if its columns form an orthonormal set.

Proof

The (i, j) entry of $P^T P$ is the dot product of column i with column j . Thus $P^T P = I$ exactly when the columns have norm 1 and are mutually orthogonal. $\square$

Orthogonality of distinct eigenspaces

For a symmetric matrix, eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof

If $A\mathbf{x} = \lambda\mathbf{x}$ and $A\mathbf{y} = \mu\mathbf{y}$, then symmetry gives

$$\mathbf{x}^T A\mathbf{y} = (A\mathbf{x})^T \mathbf{y}.$$

The left side is $\mu\mathbf{x}^T \mathbf{y}$ and the right side is $\lambda\mathbf{x}^T \mathbf{y}$. Therefore

$$(\lambda - \mu)\mathbf{x}^T \mathbf{y} = 0.$$

For $\lambda \neq \mu$, $\mathbf{x}^T \mathbf{y} = 0$. $\square$

Reality of eigenvalues

Every eigenvalue of a real symmetric matrix is real.

Proof

Allow a complex eigenvector $\mathbf{z} \neq 0$. From $A\mathbf{z} = \lambda\mathbf{z}$,

$$\mathbf{z}^* A\mathbf{z} = \lambda \mathbf{z}^* \mathbf{z}.$$

The denominator $\mathbf{z}^* \mathbf{z}$ is positive real. Moreover,

$$\overline{\mathbf{z}^* A\mathbf{z}} = \mathbf{z}^* A^T \mathbf{z} = \mathbf{z}^* A\mathbf{z},$$

so the numerator is real. Hence

$$\lambda = \frac{\mathbf{z}^* A\mathbf{z}}{\mathbf{z}^* \mathbf{z}}$$

is real. $\square$

Spectral theorem

A real matrix is orthogonally diagonalizable if and only if it is symmetric.

Proof framework

If $A = PDP^T$, then $A^T = PD^T P^T = A$. Conversely, for symmetric A , choose a unit eigenvector $\mathbf{q}_1$. The orthogonal complement $\mathbf{q}_1^\perp$ is invariant because

$$\mathbf{q}_1^T A \mathbf{y} = (A \mathbf{q}_1)^T \mathbf{y} = \lambda_1 \mathbf{q}_1^T \mathbf{y} = 0.$$

Restrict A to this invariant subspace and apply induction. Combining the resulting orthonormal eigenbasis with $\mathbf{q}_1$ yields an orthonormal eigenbasis for the whole space. $\square$

Detailed orthogonal diagonalization

For

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix},$$

$$\det(A - \lambda I) = (5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1).$$

For $\lambda = 6$,

$$A - 6I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix},$$

so choose $\mathbf{v}_1 = (2, 1)$. For $\lambda = 1$,

$$A - I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix},$$

so choose $\mathbf{v}_2 = (-1, 2)$. Their dot product is

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = 2(-1) + 1(2) = 0.$$

Both norms equal $\sqrt{5}$. Normalize:

$$\mathbf{q}_1 = \frac{1}{\sqrt{5}}(2, 1), \quad \mathbf{q}_2 = \frac{1}{\sqrt{5}}(-1, 2).$$

Then

$$P = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Check orthogonality explicitly:

$$P^T P = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I.$$

Finally,

$$P^T A P = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}.$$

For

$$C = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix},$$

the eigenvalue 1 has a two-dimensional eigenspace. One basis is

$$\mathbf{v}_1 = (1, -1, 0), \quad \mathbf{v}_2 = (1, 0, -1).$$

These vectors are not orthogonal because $\mathbf{v}_1 \cdot \mathbf{v}_2 = 1$. Apply Gram–Schmidt:

$$\begin{aligned}\mathbf{w}_1 &= \mathbf{v}_1, \\ \mathbf{w}_2 &= \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1 \\ &= (1, 0, -1) - \frac{1}{2}(1, -1, 0) \\ &= \left(\frac{1}{2}, \frac{1}{2}, -1\right).\end{aligned}$$

Multiplying by 2 gives the simpler orthogonal vector $(1, 1, -2)$. After normalization,

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \mathbf{q}_2 = \frac{1}{\sqrt{6}}(1, 1, -2).$$

For eigenvalue 4, choose

$$\mathbf{q}_3 = \frac{1}{\sqrt{3}}(1, 1, 1).$$

These three vectors form an orthonormal eigenbasis, and the resulting orthogonal matrix satisfies

$$P^T C P = \text{diag}(1, 1, 4).$$

4 Applications

Stable age distribution

Use the fictional hiking-preserve wildlife matrix

$$L = \begin{bmatrix} 0 & 6/5 & 2 \\ 11/20 & 0 & 0 \\ 0 & 11/20 & 0 \end{bmatrix}.$$

Then

$$L \begin{bmatrix} 20 \\ 10 \\ 5 \end{bmatrix} = \frac{11}{10} \begin{bmatrix} 20 \\ 10 \\ 5 \end{bmatrix}.$$

Thus the growth factor is 1.1, and the stable age proportions are 4/7, 2/7, and 1/7.

A diagonalized differential system

Consider

$$\mathbf{y}' = A\mathbf{y}, \quad A = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(A - \lambda I) = (-3 - \lambda)^2 - 1 = (\lambda + 2)(\lambda + 4).$$

Thus the eigenvalues are -2 and -4 , with eigenvectors $(1, 1)$ and $(1, -1)$. Form

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}.$$

Use the substitution $\mathbf{y} = P\mathbf{w}$. Since $\mathbf{y}' = P\mathbf{w}'$, the original equation becomes

$$P\mathbf{w}' = AP\mathbf{w}.$$

Multiplying by P^{-1} gives

$$\mathbf{w}' = P^{-1}AP\mathbf{w} = D\mathbf{w}.$$

Therefore

$$w_1' = -2w_1, \quad w_2' = -4w_2,$$

with solutions

$$w_1 = c_1 e^{-2t}, \quad w_2 = c_2 e^{-4t}.$$

Returning to the original variables,

$$\mathbf{y}(t) = c_1 e^{-2t} (1, 1)^T + c_2 e^{-4t} (1, -1)^T.$$

If $\mathbf{y}(0) = (3, 1)^T$, then

$$(3, 1) = c_1 (1, 1) + c_2 (1, -1),$$

so

$$c_1 + c_2 = 3, \quad c_1 - c_2 = 1.$$

Adding gives $c_1 = 2$, and then $c_2 = 1$. Hence

$$\boxed{\mathbf{y}(t) = 2e^{-2t} (1, 1)^T + e^{-4t} (1, -1)^T}.$$

Quadratic forms and principal axes

A quadratic form

$$q(x, y) = ax^2 + bxy + cy^2$$

has matrix

$$A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}.$$

Principal-axes theorem

If $P^T A P = D$ with P orthogonal and $\mathbf{x} = P\mathbf{u}$, then

$$q(\mathbf{x}) = \mathbf{u}^T D \mathbf{u}.$$

Thus all cross terms disappear.

Proof

$$q = (P\mathbf{u})^T A (P\mathbf{u}) = \mathbf{u}^T P^T A P \mathbf{u} = \mathbf{u}^T D \mathbf{u}.$$

Because D is diagonal, only squared terms remain. $\square$

For

$$5x^2 + 4xy + 2y^2 = 12,$$

use

$$A = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}, \quad P = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Use the coordinate change

$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} u \\ v \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2u - v \\ u + 2v \end{bmatrix}.$$

Then

$$q(x, y) = \begin{bmatrix} u & v \end{bmatrix} P^T A P \begin{bmatrix} u \\ v \end{bmatrix} = 6u^2 + v^2.$$

Hence the rotated equation is

$$6u^2 + v^2 = 12, \quad \frac{u^2}{2} + \frac{v^2}{12} = 1.$$

This is an ellipse with semiaxis lengths $\sqrt{2}$ and $2\sqrt{3}$. The principal axes are the two eigenvector directions.

Classification of quadratic forms

For a symmetric matrix A , the signs of the eigenvalues determine the type of quadratic form:

Eigenvalue signs	Classification
All positive	Positive definite
All negative	Negative definite
Both positive and negative	Indefinite
Nonnegative with at least one zero	Positive semidefinite
Nonpositive with at least one zero	Negative semidefinite

Constrained optimization

Extrema of a quadratic form on the unit sphere

For symmetric A , the maximum of $\mathbf{x}^T A \mathbf{x}$ subject to $\|\mathbf{x}\| = 1$ is the largest eigenvalue, and the minimum is the smallest eigenvalue.

Proof

Write $A = PDP^T$ and set $\mathbf{y} = P^T \mathbf{x}$. Orthogonality gives $\|\mathbf{y}\| = 1$. Then

$$\mathbf{x}^T A \mathbf{x} = \sum_{i=1}^n \lambda_i y_i^2, \quad \sum_{i=1}^n y_i^2 = 1.$$

This is a weighted average of the eigenvalues, so it lies between the smallest and largest eigenvalues. Equality occurs at the corresponding unit eigenvectors. $\square$

For $5x^2 + 4xy + 2y^2$ on $x^2 + y^2 = 1$, the maximum is 6 at $\pm(2, 1)/\sqrt{5}$, and the minimum is 1 at $\pm(-1, 2)/\sqrt{5}$.

Chapter terminology

Eigenvalue	Scalar describing the scaling of an eigenvector.
Eigenvector	Nonzero vector satisfying $A\mathbf{v} = \lambda\mathbf{v}$.
Eigenspace	Nullspace $\ker(A - \lambda I)$.
Characteristic polynomial	$\det(A - \lambda I)$.
Algebraic multiplicity	Multiplicity of an eigenvalue as a polynomial root.
Geometric multiplicity	Dimension of the eigenspace.
Diagonalization	Factorization $A = PDP^{-1}$.
Symmetric matrix	Matrix satisfying $A^T = A$.
Orthogonal matrix	Matrix satisfying $P^T P = I$.
Spectral theorem	Symmetric matrices possess orthonormal eigenbases.
Stable age distribution	Positive eigenvector giving long-run age proportions.
Quadratic form	Expression $\mathbf{x}^T A \mathbf{x}$.
Principal axes	Orthogonal eigenvector directions of a quadratic form.
Rayleigh quotient	$\mathbf{x}^T A \mathbf{x} / (\mathbf{x}^T \mathbf{x})$.